Math 372: Practice Midterm 2

Problem 1. An evil sorcerer decides to summon some minions to help clean up his lawn after a big storm. His magic spell summons up to 2 skeletons and up to 2 abyssal specters. Let X be the number of skeletons summoned, and let Y be the number of abyssal specters summoned. According to the notes in his spellbook, the joint pmf of X and Y is given by:

$p(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	0.12	0.10	0.08
$x=1$	0.06	0.18	0.14
$x=2$	0.04	0.08	0.20

(a) What is the probability that the necromancer summons an even number of minions?

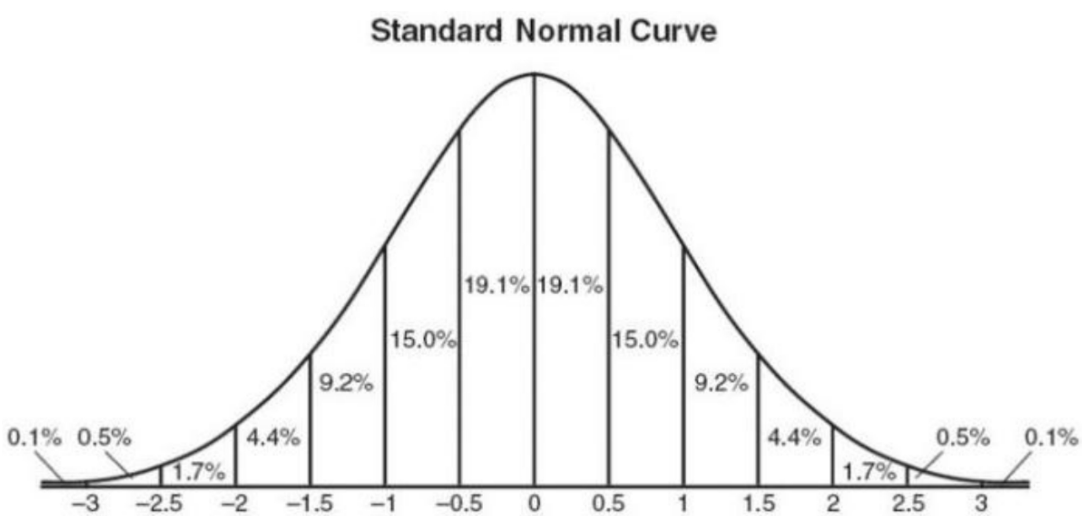
(b) Compute $\mathbb{P}[X \leq 1 \text{ and } Y \leq 1]$.

(c) Using words, describe the event $[X+Y \geq 2]$, and then compute its probability.

(d) Compute the marginal pmf of X and Y. Using $p_X(x)$, find $\mathbb{P}[X \geq 1]$.

(e) Are X and Y independent? Justify your answer.

Problem 2. A machine that produces ball bearings has initially been set so that the true average diameter of the bearings it produces is .500 cm. A bearing is acceptable if its diameter is within .004 cm. of this target value. Suppose, however, that the setting has changed during the course of production, so that the bearings have normally distributed diameters with mean value .499 cm. and standard deviation .002 cm. What percentage of the bearings produced will not be acceptable? [Hint: use the following plot of the standard normal density curve.]



Problem 3. Fix $\beta > 0$, and let X be a random variable with cdf

$$F(t) = \begin{cases} 1 - \left(\frac{\beta}{t}\right)^3 & : \quad t \geq \beta \\ 0 & : \quad t < \beta \end{cases}$$

(a) Find f, the pdf of X.

(b) Find $\mathbb{E}[X]$.

(c) Find $\mathbb{P}[\beta \leq X \leq 2\beta]$.

Problem 4. Let T be the time, in minutes, between successive drink orders at a bar. Suppose that a reasonable probability model for T is given by the pdf

$$f_T(t) = Ce^{-\frac{t}{4}}\mathbf{1}_{[t\geq 0]},$$

where is $C > 0$ is some number that you'll have to determine.

(a) Find a value for C such that $f_T(t)$ is a valid pdf.

(b) What is the probability that the time between successive orders is greater than 4 minutes?

(c) On average, how long must the bartender wait between drink orders?

Problem 5. It is a well-known fact that 1% percent of treasure chests are actually mimics.[1] A wizard has developed a magical detection spell to identify mimics before opening them. However, the spell is imperfect:

$$\mathbb{P}\left[\text{spell says ``safe''} \mid \text{chest is a mimic}\right] = 0.03$$

$$\mathbb{P}\left[\text{spell says ``mimic''} \mid \text{chest is normal}\right] = 0.05$$

(a) What is the probability that a randomly-selected treasure chest chest is flagged as a mimic?

(b) Suppose a treasure chest is flagged as a mimic. What is the probability that it is actually a mimic?

Problem 6. Let A and B be events for which you know $\mathbb{P}[A \cup B] = 0.8$, $\mathbb{P}[A] = 0.4$, and $\mathbb{P}[A \cap B] = 0.3$.

(a) Find $\mathbb{P}[A]$

(b) Are A and B independent? Justify your answer.

(c) Are A and B disjoint? Justify your answer.

(d) Find $\mathbb{P}[A \cap B^c]$

(e) Find $\mathbb{P}[A \mid A \cup B]$

(f) Find the probability that exactly one of the two events A, B occurs.

Problem 7. Consider a deck of four cards consisting of one card numbered '1', two cards numbered '2', and one card numbered '3'. The deck is suffled thoroughly. Let D denote the difference between the numbers on the top two cards, ignoring the sign.

(a) Display the probability mass function of D in a suitable table.

(b) Find $\mathbb{E}[D]$

(c) Find $\mathrm{Var}(D)$

[1] A *mimic* is a type of monster which disguises itself as a treasure chest and attempts to eat you when you open it.